

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457924.9193 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.066
Transit depth (Rp/Rs)^2	4.65 +/- 2.84 [%]
Orbital Inclination (inc)	89.5 +/- 2.77
Airmass coefficient 1 (a1)	0.27 +/- 0.17
Airmass coefficient 2 (a2)	0.45 +/- 0.11
Scatter in the residuals of the lightcurve fit is	50.95 %
Best Comparison Star	#3 - [143, 457]
Optimal Aperture	10.84
Optimal Annulus	27.66
Transit Duration (day)	0.0875 +/- 0.0083

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.066 vs 0.1646 (deviation 0.71σ)
Duration fit vs published	0.0875 d vs 0.0758 d (deviation 1.28σ)
Mid-time O–C	5.9 min
Depth SNR	3.0
Residual scatter	50.95 %

Light curve

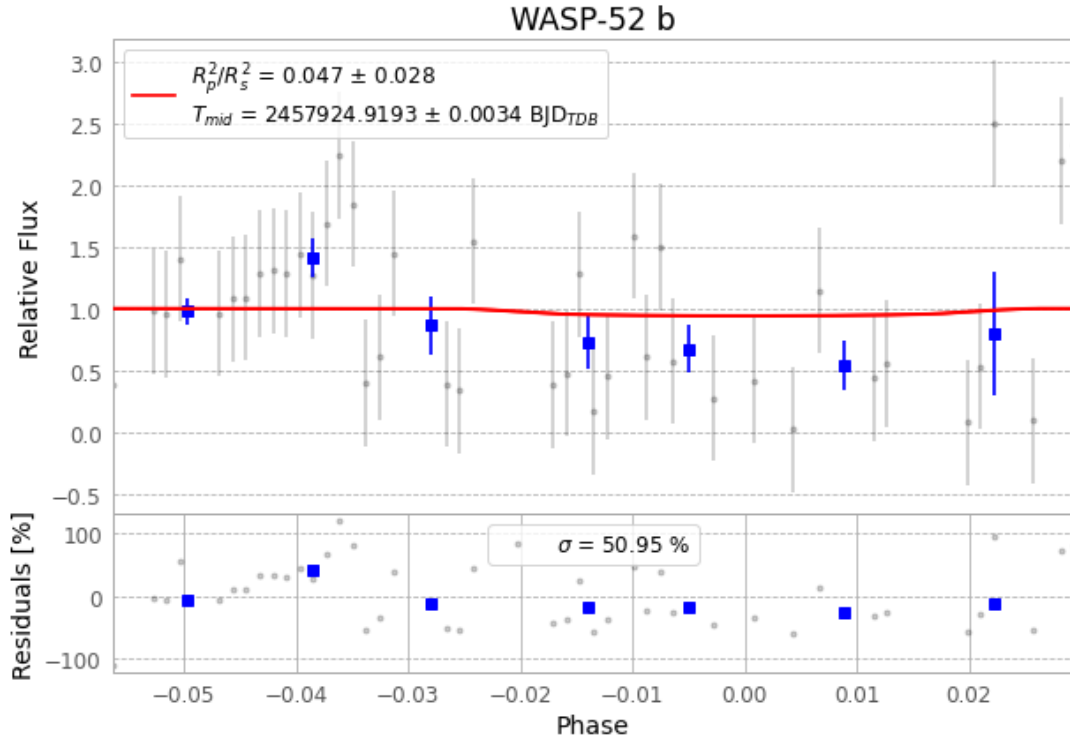


Figure 1 — FinalLightCurve_WASP-52 b_2017-06-20.png (SHA-256 94f3b642296e70aa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [551, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 41dd53ce793a2e95...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 72f1d3a

Data provenance

73 FITS frames (48 MB), WASP-52170620073912.FITS → WASP-52170620111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-170620.log (ce54667c9a28...), FinalLightCurve_WASP-52 b_2017-06-20.png (94f3b642296e...), FinalLightCurve_WASP-52 b_2017-06-20.csv (f239e9e3c5b9...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 11:59Z.